Carina Seitz

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University of Nevada, Reno | Tahoe Institute for Global Sustainability | Department of Biology

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CURRENT APPOINTMENTS

Research Assistant Professor. Department of Biology, Tahoe Institute for Global Sustainability, University of Nevada, Reno, USA. **2024-Present**.

- Investigate various types of particles impacting Lake Tahoe's clarity
- Design and conduct experiments to assess Golden mussel survival and reproduction in Lake Tahoe, Mead Lake, and Pyramid Lake
- Develop a pipeline for raw data collection, cleaning, structuring, and software development for watershed observatory networks at Summit Lake (SWEON) and Lake Tahoe (TEON)

Research Scientist. Departamento de Geología y Petróleo, Centro Regional Universitario Bariloche (CRUB) (Universidad Nacional del Comahue-CONICET), Argentina. **2025-Present**.

- Limnological and Paleolimnological studies of aquatic ecosystems with emphasis on Patagonia.

PREVIOUS PROFESSIONAL APPOINTMENT

Research Assistant. Departamento de Geología y Petróleo, Centro Regional Universitario Bariloche (CRUB) (Universidad Nacional del Comahue-CONICET), Argentina. 2022-2025

- Limnological and Paleolimnological studies of aquatic ecosystems, with an emphasis on Patagonia.

Lecturer, Department of Geology and Petroleum, Centro Regional Universitario Bariloche (CRUB), Universidad Nacional del Comahue, Argentina. 2023- 2024

- Teaching undergraduate courses in Biology, Petroleum and Civil engineering majors.

Lecturer. Department of Geology, Universidad Nacional del Sur, Argentina. 2015-2023

- Teaching undergraduate courses in Biology and Geology majors

Postdoctoral fellow CONICET - Instituto Argentino de Oceanografia - National Council of Scientific and Technical Research (IADO-CONICET), Argentina. 2019-2022

- Ecological regime shift on Pampean Shallow Lakes.

Postdoctoral Researcher. Department of Biology- Global Water Center, University of Nevada, Reno, USA. 2021-2021

- Limnological and paleoecological studies on Mountain Lakes.
- Experiments on quagga mussels' survival in changing physical and chemical conditions.

Postdoctoral Fellow. Queen Elizabeth II Advanced Schollar. Department of Geology, University of Regina (SK, Canada). 2020

- Paleoclimate record as a tool to understand the effect of future climate change on shallow lakes.

Lead Geologist at environmental consultant firm Auditora 2000 S.A., Argentina. 2010-2013

- Environmental monitoring and impact assessment
- Leader in the management of hazardous wastes in Rally Dakar 2013 Peru-Argentina-Chile

EDUCATION

PhD Geology, Universidad Nacional del Sur (Argentina). 2019

Dissertation: "Geological, geomorphological and limnological evolution of Pampean lakes in the southwestern Buenos Aires province"

B.S. Geology. Universidad Nacional de Rio Cuarto (UNRC) (Argentina). 2009

Honors thesis: "Neotectonic study of new sections of the Las Lagunas fault, Southern Córdoba Province, Argentina".

PEER-REVIEWED PUBLICATIONS (Google Schoolar: *h-index: 15 | i10-index: 19*)

(Students, or technicians mentored or co-mentored by Seitz are underlined)

37. Flynn, K., Simmons, J., Strachan, S., Chandra, S., Seitz, C., Savran, W., ... & Heggli, M. (2026). The Summit Lake Watershed Environmental Observation Network: A Design Framework for a High-Throughput, Remote Field Sensor Network. In *Proceedings of the Practice and Experience in Advanced Research Computing 2026: Resilient Roots+ Empowered Communities*.10.1145/3785462.3815808

36. **Seitz C**, Leira M, Rodríguez-Gómez CF, Castro-Cera L, Vouilloud AA, Chamorro S, Blanco S, Goldenberg-Vilar A, De Gracia J, Delgado C, Sepúlveda-Zúñiga EA, Pérez L, Lopera Congote L, Romero LE, Cuña-Rodríguez C, Novelo E, Baffico G, Trobajo R. 2025. DIASPORE: an open bibliographic dataset of diatom research in Spanish-speaking countries (1844–2024). *Diatom Research*, 1-9. https://doi.org/10.1080/0269249X.2026.2710064

36. Alfonso, M. B., Scordo, F., Seitz, C., Arias, A. H., Ronda, A. C., Mavo Manstretta, G. M., ... & Piccolo, M. C. 2026. Microplastic contamination in remote mountain lakes of the americas: a baseline assessment from Patagonia and Northern California. *Frontiers in Toxicology*. 10.3389/ftox.2026.1851245

35. Tromboni F, Jativa C, **Seitz C**, Maasri A, Mohr S, Grossart H-P, Grandi G, Bertuzzo E, Jähnig SC, Mendoza-Lera C, Lorke A, Cantonati M, Lupon A, Bernal S. 2025. Flow and Biodiversity Shifts Drive Riverine Ecosystem Metabolism: Insights from Experimental Stream Mesocosms. Limnology and Oceanography. http://doi.org/10.1002/lno.70285

34. Scordo F, **Seitz C**, Buria L, Piccolo MC, Perillo GME. 2025. Climatic and Human Impacts in Patagonian Terminal Lakes. Science of Total Environment, 966. https://doi.org/10.1016/j.scitotenv.2025.178714

33. Córdoba F, **Seitz C**, Pisani N, Santamans CD, Guerra L, Navarro D, Stutz S, Piovano E. 2025. Chapter 18 Paleolimnological Records in the Southwestern Pampa Plain. In Pampean Lakes, Stutz S and Piovano E (Editors). Springer Nature. Hardcover ISBN 978-3-031-86027-0. https://link.springer.com/book/9783031860270

32. San Pedro AP, Scordo F, **Seitz C**, Krynak EM, Girdner SF, Blaszczak J, Chandra S 2025. Context-dependent controls of periphyton across the littoral-benthic habitat of deep, clear lakes. Aquatic Sciences. https://doi.org/10.1007/s00027-025-01215-w

31. Scordo F, **Seitz C**, Abasto B, Spetter CV, Piccolo MC, Buria L, Perillo GME. 2025. Nutrients limiting the growth of planktonic and benthic-littoral primary producers in Patagonian Andean Lakes. Inland Waters, 1–31. https://doi.org/10.1080/20442041.2024.2418681

30. Scordo F, **Seitz C**, Suenaga E, Piccolo MC, Chandra S, Amodeo M, Perillo GME. 2024 An inexpensive method for the measurement of photosynthetically active radiation profiles in waterbodies. Aquatic Sciences 86:68. https://doi.org/10.1007/s00027-024-01082-x

29. Scordo F, **Seitz C**, Abasto B, Spetter CV, Piccolo MC, Chandra S, Perillo GME. 2024. Habitat-specific metabolism and nutrient limitation within an oligotrophic Patagonian Lake. Hydrobiologia. https://doi.org/10.1007/s10750-023-05449-3

28. Bess Z, Koning A, Simmons J, Suenaga E, San Pedro A, Culpepper J, Scordo F, **Seitz C**, Rhoades S, McKinnon T, McKim R, Feher K, Tromboni F, Regan JW, Chandra S. 2024. Fish Diversity and use of Nearshore and Open-Water Habitats in Terminal Lakes. Fisheries. http://dx.doi.org/10.1002/fsh.11028

27. Alfonso MB, Scordo F, Epele L,…, **Seitz C** et al. 2024. Filling the Gap a Comprehensive Freshwater Network to Map Microplastics across Ecological Gradients in Argentina. Bulletin Limnology and Oceanography, ASLO. https://doi.org/10.1002/lob.10641

26. **Seitz C**, Scordo F, Suenaga E, Carlson EM, McMillen S, Gregory L, Chandra S. 2023. Salinity and pH effects on survival, growth, and reproduction of quagga mussels. PeerJ. https://doi.org/10.7717/peerj.15450

25. **Seitz C**, Kenney WF, Patterson-Boyarski B, Curtis JH, Velez MI, Glodzik K, Escobar J, Brenner M. 2023. Sea level changes and paleoenvironmental responses in a coastal Florida salt marsh over the last three centuries. Journal of Paleolimnology. https://doi.org/10.1007/s10933-022-00275-4

24. Scordo F, Girdner SF, San Pedro A, **Seitz C**, Chandra S. 2023. Deterioration of the Littoral–Benthic Ecosystem Following Recent Expansion of Signal Crayfish (Pacifastacus leniusculus) in the World's Clearest Large Lake. Ecosystems https://doi.org/10.1007/s10021-023-00846-0

23. Noble P, **Seitz C**, Lee S, Manoylov KM, Chandra S. 2023. Characterization of algal community composition and structure from the nearshore environment, Lake Tahoe (United States). Frontiers in Ecology and Evolution. https://doi.org/10.3389/fevo.2022.1053499

22. Kulkarni C, Jara I.A., Chevalier M., Benito X,…, **Seitz C**, et al. 2023. pSESYNTH project: Community mobilization for a multi-disciplinary paleo database of the Global South. Pages magazine, volume 31 (1). special section: advancing past socio–environmental systems science. https://doi.org/10.22498/pages.31.1.30

21. Scordo F, **Seitz C**, Fiorenza JE, Piccolo MC, Perillo GME. 2023. Human impact changes hydrological connectivity in a Patagonian fluvial basin. Journal of Hydrology: Regional Studies. https://doi.org/10.1016/j.ejrh.2023.101315

20. **Seitz C**, Vélez MI, Perillo GME. 2022. Response of shallow lakes in the arid-semiarid Pampas of Argentina to Late Holocene hydroclimatic change. Quaternary International. https://doi.org/10.1016/j.quaint.2021.09.016

19. Scordo F, Lottig NR, Fiorenza JE, Culpepper J, Simmons J, **Seitz C**, et al. 2022. Hydroclimate variability affects habitat-specific (open water and littoral) lake metabolism. Water Resources Research, 58, e2021WR031094. https://doi.org/10.1029/2021WR031094

18. Scordo F, Sadro S, Culpepper J, **Seitz C**, Chandra S. 2022. Wildfire Smoke Effects on Lake-Habitat Specific Metabolism: Toward a Conceptual Understanding. Geophysical Research Letters, 49, e2021GL097057. https://doi.org/10.1029/2021GL097057

17. Toniolo MA, **Seitz C**, Perillo GME. 2021 Origin and evolution of tidal depressions in a tidal flat and their role in carbon sequestration in the Bahía Blanca Estuary (Argentina). Marine Geology. https://doi.org/10.1016/j.margeo.2021.106467

16. Scordo F, Chandra S, Suenaga E, Kelson JS, Culpepper J, Scaff L, Tromboni F, Caldwell TJ, **Seitz C**, Fiorenza JE, Williamson CE, Sadro S, Rose KC, Poulson SR. 2021. Smoke from regional wildfires alters lake ecology. Scientific Report. https://doi.org/10.1038/s41598-021-89926-6.

15. **Seitz C**, Scordo F, Vitale AJ, Velez MI, Perillo GME. 2020. The effects of extreme climatic events on the morphometry of shallow lakes: implications for sediment resuspension and littoral and pelagic zone distribution. Journal of South American Earth Science. https://doi.org/10.1016/j.jsames.2020.102743

14. Scordo F, Spetter CV, **Seitz C**, Piccolo MC, Perillo GME. 2020. Spatial and seasonal dynamics of water physical-chemical parameters in rivers and lakes of an Argentinian Patagonia basin. Environmental Earth Sciences, 79, 322. https://doi.org/10.1007/s12665-020-09063-7

13. Scordo F, **Seitz C**, Melo WD, Piccolo MC, Perillo GME. 2020. Natural and human impacts on the landscape modeling and hydrography of the Chico River basin (Argentinean Patagonia). Catena. https://doi.org/10.1016/j.catena.2020.104783

12. López Blanco C, Rodríguez-Abaunza GA, **Seitz C**, Pérez L, Cuña-Rodriguez C, Fontana SL. 2020. Hydroclimatic variability in the Pampean Lagoon Laguna Blanca Grande (Buenos Aires, Argentina) inferred from a multi-proxy record. In Journal of South American Earth Science. https://doi.org/10.1016/j.jsames.2020.103000

11. Genchi SA, Vitale AJ, Perillo GME, **Seitz C**, Delrieux CA. 2020. Mapping Topobathymetry in a Shallow Tidal Environment Using Low-Cost Technology. Remote Sensing 12, 1394, 17p. https://doi.org/10.3390/rs12091394

10. Alfonso MB, Scordo F, **Seitz C**, Mavo Manstretta GM, Ronda A., Arias A.H., Tomba JP, Silva LI, Perillo GME, Piccolo MC. 2020. First evidence of microplastics in andean and extra-andean Patagonian lakes from Argentina. Science of the Total Environment. 8 p. https://doi.org/10.1016/j.scitotenv.2020.139385

9. **Seitz C**, Velez MI, Perillo GME. 2019. Cenozoic Geologic Evolution of The Lower Colorado River Basin, Northern Patagonia, Argentina. Andean Geology. 46 (1): 131-152. http://dx.doi.org/10.5027/andgeoV46n1-3060

8. Zilio M, **Seitz C**, Scordo F, Gil V, Zapperi P, Costilla P, Huamantinco Cisneros MA, Perillo GME, Piccolo MC. 2018. Is Collaborative Management Always Possible? The Case Of Sauce Grande River Basin, Argentina. The International Journal of River Basin Management. 27p. https://doi.org/10.1080/15715124.2018.1546727

7. Vélez MI, Conde D, Lozoya JP, Rusak JA, Garcia-Rodriguez F, **Seitz C**, Harmon T, Perillo GME, Escobar J, Vilardy SP. 2018. Paleoenvironmental Reconstructions Improve Ecosystem Services Risk Assessment: Case Studies From Two Coastal Lagoons In South America. Water. 18p. https://doi.org/10.3390/w10101350

6. Scordo F, **Seitz C**, Melo WD, Piccolo MC, Perillo GME. 2018. Assessment of geomorphological and hydrological changes produced by Pleistocene glaciations in a Patagonian basin. Journal of South American Earth Sciences 83 (2018) 195-209p. https://doi.org/10.1016/j.jsames.2018.03.001

5. Scordo F, Lavender TM, **Seitz C**, Perillo VL, Rusak JM, Piccolo MC, Perillo GME. 2018. Modeling aquatic ecosystem services: assessing the role of site and region-specific attributes in determining model performance. Water. 42p. https://doi.org/10.3390/w10111496

4. Alfonso MB, Brendel AS, Vitale AJ, **Seitz C**, Piccolo MC, Perillo GME. 2018. Drivers of ecosystem metabolism in two managed shallow lakes with different salinity and trophic conditions: Sauce Grande and La Salada Lakes (Argentina). Water. 15p. https://doi.org/10.3390/w10091136

3. Scordo F, **Seitz C**, Zilio M, Melo WD, Piccolo MC, Perillo GME. 2017. Evolución de los Recursos Hídricos en el "Bajo de Sarmiento" (Patagonia Extra Andina): Impactos Naturales y Antrópico. Anuário do Instituto de Geociências, Vol 40 (2): p. 106-117. https://doi.org/10.11137/2017_2_106_117

2. **Seitz C,** Scordo F, Zapperi P. 2016. Estudio Geomorfológico De Chamical (Provincia De La Rioja, Argentina) Como Base Para El Análisis De Riesgos Naturales. Revista Boletín Geográfico (ISSN 0326-1735). Boletín geográfico. Año XXXVII. N°38 - 2016, pp. 61 – 78. http://170.210.83.53/htdoc/revele/index.php/geografia/article/view/1364/pdf

1. Sagripanti G., Villalba D, Bettiol A, **Seitz C**. 2009. Sismicidad y paleosismicidad en la región sur de la provincia de Córdoba. Análisis del peligro y riesgo sísmico asociado. Revista ASAGAI Nº 23: 45-53. ISSN 1851-7838.

NON-REFEREED PUBLICATIONS, REPORTS, & DATA RELEASES

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13. Tromboni F., Jativa C., Seitz C., et al. 2026. Supporting information for: Flow variability and macroinvertebrates jointly regulate stream periphyton and metabolism: insights from experimental stream mesocosms; John Wiley & Sons; https://doi.org/10.1002/lno.70285

12. Chandra S, Suenaga E, **Seitz C**, Senft K, Carlson EM, Bess Z, Scordo F, Lam M. 2025. Zooplankton Ecology of Lake Tahoe: Composition, Migration, and Influence on Plankton Particle Sizes. Final Report submitted to the Tahoe Science Advisory Council for the Tahoe Regional Planning Agency, Contract #22C00021 / Council Work Order #115 (January 2).

11. Scordo F, **Seitz C** 2025. Code & Data "Climatic and Human Impact in Patagonian Terminal Lakes Data", Mendeley Data, V1, doi: 10.17632/s3x7n67xhp.1

10. Scordo F, **Seitz C**, Abasto B, et al. 2025. Code & Data "Nutrients limiting the growth of planktonic and benthic-littoral primary producers in Patagonian Andean Lakes", Mendeley Data, V1, doi: 10.17632/htff8jhyv6.1

9. Pérez A, **Seitz C**, Scordo F, Baffico G, Temporetti P. 2024 Preliminary limnological and paleolimnological characterization of Laguna Verde, Reserva Selva Triste, Villa La Angostura (Neuquén). Submitted to the Dirección de Medio Ambiente, Municipalidad de Villa La Angostura.

8. Scordo F, **Seitz C**, Abasto B, Spetter C, Piccolo MC, Perillo GME. 2023. National Parks Report, Project 1644 – Dirección Regional Patagonia Norte. Interim report of activities conducted between November 2022 and February 2023 in lakes Gutierrez, Mascardi, Guillelmo, Steffen, Huala Hue, and Puelo.

7. Seitz C, Chandra S, Scordo F. 2022. Invasive Quagga Mussel: Potential Survival, Growth, and Reproduction in the Pyramid–Truckee Ecosystem. United States Department of Interior, Bureau of Indian Affairs, Grant No. A20AV00856, Aquatic Invasive Species Program.

6. Scordo F, **Seitz C**, Abasto B, et al. 2023. Climatic and Anthropogenic Influences on Lakes in Argentine Patagonia: Toward Consensus-Based Solutions with Social Stakeholders and Decision-Makers. Technical report # 1644 - DRPN.

5. Chandra S, Suenaga E, **Seitz C**, Carlson E, Scordo F, Noble P. 2022. The Role of Crayfish and Invertebrate Food Web Dynamics in Controlling Algae in Lake Tahoe. Final Technical Report for Agreement #18-047-160.

4. Chandra S, Scordo F, Suenaga E, Blazsczak J, **Seitz C,** et al.2022. Impacts of Smoke–Ash from the 2021 Wildfires on the Ecology of Lake Tahoe. Tahoe Science Advisory Council, TRPA Contract #17C00018.

3. Scordo F, **Seitz C**, Piccolo MC, Perillo GME. 2022. Temporal evolution of the surface area of Cronómetro Lake and vegetation dynamics (May–June 2022). STAN P 9855 and 9856–22, IADO–CONICET–UNS.

2. **Seitz C**, Vélez MI, Perillo GME. 2021. Response of shallow lakes to climate change during the last 1200 yr dataset, Mendeley Data, V2, doi: 10.17632/kb26jff2gx.2

1. Sagripanti GL, Bettiol A, **Seitz C**. 2007. *Terremotos: nuestro planeta vibra bajo el poder de su energía*. Agencia Córdoba Ciencia. Area de Promocion Cientifica de la Agencia Córdoba Ciencia SE. ISBN: 978-987-1353-03-3

PUBLICATIONS – submitted, in review, or in revision
(Students, or technicians mentored or co-mentored by Seitz are underlined)

Scordo F, Simmons J, **Seitz C**, Chandra S. 2025. Hydroclimate variability affects habitat-specific (pelagic and littoral-benthic) metabolism in a eutrophic terminal lake. Inland Waters (*in review*)

Scordo F., **Seitz, C.,** Chuter-Davies, W. et al. 2026. From Burned Watersheds to Smoke Exposure: Long-Term Wildfire Impacts on North American Lakes. Nature CEE *(in review)*

Scordo F, Redding C, Chuter-Davies W, **Seitz C**, et al. 2026. *Health impacts of wildfire emissions on North American populations* (Manuscript submitted for publication). GeoHealth.

Mast, Elizabeth ; Nussear, Kenneth ; Suenaga, Erin ; Scordo, Facundo ; Seitz, Carina ; Chandra, Sudeep ; Blaszczak, Joanna. 2026. Climate driven variation in ice-out impacts zoobenthos biodiversity and production in a subalpine lake. Ecosphere.

RESEARCH GRANTS

Total: $1,130,580

Tahoe Regional Planning Agency and Tahoe Science Advisory Council. 2026-2027. Co-PI. An integrated monitoring program to quantify the type of plankton and fate of particles controlling Lake Tahoe's clarity.	$530,000
Nevada Department of Wildlife. 2026-2027. Co-PI. Invasive Golden Mussel Survival, Growth and Reproductive Viability Assessment for Lakes Tahoe, Mead, and Pyramid Lake.	$200,000
National Scientific and Technical Research Council (Argentina). PIP 2025-2027. PI. "High-frequency temporal monitoring of pelagic and littoral zones of Patagonian lakes using sensors".	$5,000
PI, "Respuesta de los lagos patagónicos a eventos geológicos, climáticos y antrópicos a través del tiempo".2022-2025. PICT-2020-SERIEA-005 48", Fondo para la Investigación Científica y Tecnológica (FONCYT, Argentina)	$900
Instituto Argentino de Oceanografía-Universidad Nacional del Sur. STAN 9855–22, 2022. PI. "Analysis of the temporal evolution of the Cronómetro Lake surface".	$3,000
Instituto Argentino de Oceanografía-Universidad Nacional del Sur. STAN 9856–22, 2022. PI. "Analysis of the temporal evolution of the Cronómetro Lake vegetation".	$2,000
co-PI, "Invasive Quagga Mussel Survival Test for Pyramid Lake" funded by The Pyramid Lake Fisheries 2021-2022. PI: Sudeep Chandra	$178,000
Project lead, "Impacts of Smoke-Ash from the 2021 Wildfires to the Ecology of Lake Tahoe" funded by Tahoe Science Advisory Council. 2021-2022. PI: Sudeep Chandra	$210,000
co-PI, "Análisis ambiental y biológico del Cenozoico tardío en el sudoeste de la provincia de Buenos Aires: relevancia de la región en el pensamiento evolutivo de Darwin" funded by Universidad Nacional Del Sur (UNS). 2020- 2023. PI: Dr. Rodrigo Tomassini.	$1680

FELLOWSHIPS & AWARDS

pSESYNTH Workshop	$ 2000
IAL-IPA 2022 Travel award	$500
Postdoctoral Fellow, National Scientific and Technical Research Council (CONICET). 2019- 2022	$12,000
Postdoctoral Fellow, Queen Elizabeth II Advanced Schollar. 2020	$24,800
Internship at the University of Regina (Sk, Canada) funded by Universidad Nacional del Sur and Sensing the Americas' Freshwater Ecosystem Risk from Climate Change (SAFER) project grand 2016	$2,400
PhD fellow of National Scientific and Technical Research Council (CONICET). 2014-2019	$60,000
Travel Award to attend Global Lake Ecological Observatory Network meetings, GLEON 2016	$2500

TEACHING EXPERIENCE

Teaching Faculty

Fundamental of Geology and Geomorphology– Centro Regional Universitario Bariloche (CRUB), Universidad Nacional del Comahue, Argentina 2023-2024

Led weekly undergraduate laboratory sessions and lectures focused on general geology for Bachelor's degree in Biology

Environmental Geology– Centro Regional Universitario Bariloche (CRUB), Universidad Nacional del Comahue, Argentina 2023-2024

Delivered lectures to undergraduate and postgraduate students Bachelor's degree in Biology and PhD in Biology and Geology

General Geology (3557)– Centro Regional Universitario Bariloche (CRUB), Universidad Nacional del Comahue, Argentina 2023-2024

Led weekly undergraduate laboratory sessions and lectures focused on general geology for Bachelor's degrees in Civil Engineering and Petroleum Engineering

Introduction to Geology – Centro Regional Universitario Bariloche (CRUB), Universidad Nacional del Comahue, Argentina 2023-2024

Led weekly undergraduate laboratory sessions and lecture focused on general geology for the Professor in Biology's major

General Geology B (4161)– Universidad Nacional del Sur, Argentina 2014- 2023

Led weekly undergraduate laboratory sessions and lecture focused on general geology for Bachelor's degree in Biology

Marine Geology (4165)– Universidad Nacional del Sur, Argentina 2015-2023

Led weekly undergraduate laboratory sessions for Bachelor's degree in Geology

Applied Marine Geology (4166)– Universidad Nacional del Sur, Argentina 2015-2023

Led weekly undergraduate laboratory sessions for Bachelor's degree in Oceanography

General Geology (4300) – Universidad Nacional del Sur, Argentina 2014-2015

Led weekly undergraduate laboratory sessions and lecture focused on general geology for Bachelor's degree in Geology

Introduction to Geology (4161)– Universidad Nacional del Sur, Argentina 2014-2015

Led weekly undergraduate laboratory sessions and lecture focused on general geology for Degree in Environmental Technician minor

Teaching Assistant

Structural Geology – Universidad Nacional de Río Cuarto, Argentina 2007-2008

Assisted the professor in undergraduate laboratory sessions

General Geochemistry – Universidad Nacional de Río Cuarto, Argentina 2007-2008

Assisted the professor in undergraduate laboratory sessions

Guest Lectures

Invited Speaker on Earth & Environmental Sciences -Advanced Geology Colloquium Series for Fall 2025. Department of Earth & Environmental Sciences, Indiana University Indianapolis (Indianapolis, USA) 2025

Guest lecture on Paleolimnology in the limnology class (BIOL 472) of Prof. Sudeep Chandra, University of Nevada, Reno (Reno, USA)	2025
Guest lecture on Paleolimnology in the limnology class of Prof. Sudeep Chandra University of Nevada, Reno (Reno, USA)	2022
Invited Speaker on Geography Research Seminars, Department of Geography and Environmental Studies, University of Regina (SK, Canada)	2020
Invited Speaker on workshop in statistics applied to paleoecology with R. University of Regina (SK, Canada)	2020
Invited Speaker on workshop in GIS and identification of extreme events on lakes, University of Regina (SK, Canada)	2020
Guest lecture to secondary school, Learning about La Salada shallow lake and its importance to preservation (every year)	2014-2018

MENTORING EXPERIENCE

Graduate Students (2)

Co-Advisor. Master student Aldo San Pedro, M.Sc. student, University of Nevada, Reno (USA). Focused on experiment design, environmental monitoring, and statistical analysis of lakes. Graduated 2024.	2023–2024
Co-Advisor. Master student Elizabeth Mast. University of Nevada, Reno. Master student. Graduated 2024. Supervisor: Dr. Sudeep Chandra.	2023-2024

Undergraduate Students (11)

Advisor. Undergraduate student Russell Dampong. PREP mentoring Fellow on data analysis training. Research project: Tahoe Environmental Observatory Network. Investigated terrestrial and nearshore connectivity and its impact on Lake Tahoe's clarity. Department of Biology, University of Nevada, Reno	2026
Advisor. Undergraduate student Olive Hamner-Jillson. Laboratory, and fieldwork training. Research project: Tahoe Environmental Enhancement Monitoring. Investigated various types of particles impacting Lake Tahoe's clarity. Department of Biology, University of Nevada, Reno	2026
Advisor. Undergraduate student Callie J Adam. Laboratory, fieldwork, and data analysis training. Research project: Investigated various types of particles impacting Lake Tahoe's clarity. Department of Biology, University of Nevada, Reno	2026
Advisor. Undergraduate student Macey Gilman. Laboratory, fieldwork, and data analysis training. Research project: Investigated various types of particles impacting Lake Tahoe's clarity. Department of Biology, University of Nevada, Reno	2026
Advisor. Undergraduate student Consuelo Del Rio. Laboratory, fieldwork, and data analysis training. Research project: Investigated various types of particles impacting Lake Tahoe's clarity. Department of Biology, University of Nevada, Reno	2025

Advisor. Undergraduate student Zoe Avery. Laboratory, fieldwork, and data analysis training. Research project: Investigated various types of particles impacting Lake Tahoe's clarity. Department of Biology, University of Nevada, Reno — 2025

Advisor. Professional internship undegraduate student Johana Aviles, Environmental Engineering, Universidad Nacional de Río Negro. Preliminary limnological and paleolimnological characterization of Laguna Verde, Reserva Selva Triste, Villa La Angostura (Neuquén, Argentina). — 2024

Advisor. Undergraduate student Shaye McMillen. Training in water chemistry analysis and mesocosm experiment. Research project: Salinity and pH effects on survival, growth, and reproduction of quagga mussels. Department of Biology, University of Nevada, Reno. — 2021

Advisor. Undergraduate student Logan Gregori. Training in water chemistry analysis and mesocosm experiment. Research project: Salinity and pH effects on survival, growth, and reproduction of quagga mussels. Department of Biology, University of Nevada, Reno — 2021

Advisor. Undergraduate student Audrey Dufresne. Training in water chemistry analysis and mesocosm experiment. Research project: Salinity and pH effects on survival, growth, and reproduction of quagga mussels. Department of Biology, University of Nevada, Reno — 2021

Advisor. Undergraduate student Andrea Alarid. Training in water chemistry analysis and mesocosm experiment. Research project: Salinity and pH effects on survival, growth, and reproduction of quagga mussels. Department of Biology, University of Nevada, Reno — 2021

Co-Supervisor. Honor thesis of Brittany Boyarski: Paleoenvironmental reconstruction and sea level variations based on diatom analysis of Cedar Key salt marsh, Florida, USA. University of Regina, Canada. Supervisor: Dr. Maria I Velez — 2020

Advisor. Undergraduate student Antonela Toniolo. Laboratory, fieldwork, data analysis, and writing training. Research project: Origin and evolution of tidal depressions in a tidal flat and their role in carbon sequestration in the Bahía Blanca Estuary, Instituto Argentino de Oceanografia (UNS-CONICET), Argentina — 2018-2019

Technicians (5)

Internship Advisor. Technical staff, Elizabeth Everest. Laboratory, fieldwork, and data analysis training. Research project: Investigated various types of particles impacting Lake Tahoe's clarity. Department of Biology, University of Nevada, Reno — 2026

Internship Advisor. Technical staff, Consuelo del Rio. Laboratory, fieldwork, and data analysis training. Research project: Investigated various types of particles impacting Lake Tahoe's clarity. Department of Biology, University of Nevada, Reno — 2026

Internship Advisor. Graduate student Sindhu Yalamati, MS student. Mentoring on database and software development for watershed observatory networks at Summit Lake (SWEON) and Lake Tahoe (TEON). Department of Biology, University of Nevada, Reno. — 2025

Internship Advisor. Graduate student Mukul Badhan, PhD student. Mentoring on database and software development for watershed observatory networks at Summit Lake (SWEON). Department of Biology, University of Nevada, Reno. — 2025

Internship Advisor. Graduate student Parker Land graduate student. Laboratory image processing of periphyton samples of Lake Tahoe. Department of Biology, University of Nevada, Reno. — 2021

PROFESSIONAL SERVICE, MEMBERSHIP, & TRAINING

Reviewer:

Hydrobiologia (3), book chapter Pampean Lakes book-Springer (1), Paleolimnology (4), Catena (1), Environmental Monitoring and Assessment (1), Boletín de la Sociedad Geológica Mexicana (1).

2026.- Ad-hoc reviewer article JOURNAL OF PALEOLIMNOLOGY

2025.- Ad-hoc reviewer article Hydrobiologia ISSN: 0018-8158. HYDR-D-25-00416_R1

2025.- Ad-hoc reviewer article Hydrobiologia ISSN: 0018-8158. HYDR-D-25-00416

2023.- Ad-hoc reviewer book chapter Pampean Lakes. Springer Nature

2023.- Ad-hoc reviewer article JOURNAL OF PALEOLIMNOLOGY

2022.- Ad-hoc reviewer Hydrobiologia ISSN: 0018-8158. HYDR-D-22-00239

2020.- Ad-hoc reviewer JOURNAL OF PALEOLIMNOLOGY ISSN: 0921-2728. JOPL-D-20-00078_reviewer

2020.- Ad-hoc reviewer JOURNAL OF PALEOLIMNOLOGY ISSN: 0921-2728. JOPL-D-20-00039_reviewer

2020.- Ad-hoc reviewer ENVIRONMENTAL MONITORING AND ASSESSMENT, ISSN: 0167-. EMAS-D-20-00510

2020.- Ad-hoc reviewer CATENA ISSN: 0341-8162.CATENA12212

2020.- Ad-hoc reviewer journal Boletín de la Sociedad Geológica Mexicana ISSN: 1405-3322. Olivares-Casillas

Professional memberships:

International Society of Limnology, Association for the Sciences of Limnology and Oceanography, Global Lake Ecosystem Observatory Network, International Society of Diatom Research, Past Global Change, and International Union of Quaternary Research.

Professional Service:

2025-Present. Member of the Tahoe Science Advisory Council water quality working group representing the University of Nevada, Reno.

2025-Present. Council member at the International Society of Diatom Research.
2022-Present. Member of the Red de Diatomólogos Hispanohablantes. Leading the publication working group.

2025. Led and facilitated international partnership discussions resulting in proposed academic and research agreements between UNR and partner institutions in Argentina (Universidad Nacional del Comahue and CONICET).

2025 Lake Tahoe Clarity Workshop. Tahoe Science Advisory Council hosted a two-day workshop with managers and scientists from May 6-7, Incline Village, NV.
2025. Represented the Tahoe Institute for Global Sustainability in institutional planning meetings at the UNR Tahoe Campus.
2025. Led and coordinated the establishment and operationalization of the Water Quality Laboratory (Lab-0214) at the Tahoe Center for Environmental Sciences, including procurement, installation, and commissioning of analytical equipment, cold-storage infrastructure, and chemical storage systems.
2022-2025. Regional Representative C-S América of Past Global Change (PAGES) network

2022-2025. Participant in the project pSESYNTH: The whole is not the sum of the parts: building a synthesis database of past human-environmental systems in the Global South (pSESYNTH). Funded by INQUA. https://psesynth-project.github.io/online/

Evaluations:

2022.- Doctoral committee of B.S. Carla Santamans. Thesis: "Reconstrucciones paleoambientales y paleoclimáticas durante el Holoceno tardío en Vilama, Andes Centrales". Department of geology, Universidad Nacional de Córdoba.

Conference Sessions Convened:

"SS062 Complexity in space and time: From Optical Properties to Ecosystem Processes". ASLO-SIL join meeting 2026. Co-Conveners: Dr. Sudeep Chandra, Dr. Warwich F. Vincent, Dr. Catherine O'Reilly

Training:

Research Security Training (Combined). University of Nevada, Reno	2026
Best Practices in Sponsored Project Management. University of Nevada, Reno	2026
KnowBe4 Security Awareness Training	2025
CytoSub Deep flow cytometer Training, CytoBouy company, Netherlands.	2025
Tools and strategies to prevent and raise awareness of violence against women and gender non-conforming individuals in the university setting.	2025
Laboratory Safety Hands-On Part 1 and 2. University of Nevada, Reno	2025
Preventing Harassment & Discrimination. University of Nevada, Reno	2025
Multivariate statistics for paleoenvironmental sciences	2022
Crestr Workshop: An Introductionto The Crest R package To Reconstruct Climate From Paleoecological Datasets	2022
The PaleoStats workshop. Geotop Research Centre, Université du Québec à Montréal, Canada. Workshop on computer-based workshops using the latest software and packages related to PaleoStats and chronologies.	2022
Ecology and Systematics of Algae: Cyanobacteria and Green Algae. University of IOWA. Instructor: Kalina Manoylov.	2021
Tools and strategies to prevent and raise awareness of violence against women and gender non-conforming individuals in the university setting. Universidad Nacional del Sur	2019

OUTREACH ACTIVITIES

Selected Outreach Activities

Participation in the 29th Annual Lake Tahoe Summit event representing University of Nevada, Reno at Lake Tahoe campus and the Tahoe Institute for Global Sustainability	2025
Collaborator in the Monster Fish exhibition at the University of Nevada, Reno at Lake Tahoe.	2025
Presented to the Summit Lake Paiute Tribe Council and people about Summit Lake Watershed Observatory Network website in Summit Lake	2025
Collaborator in the Institute for Research on Biodiversity and Environment (Argentina) Open Program for the General Public.	2024

Created and delivered the bathymetric map of Laguna Blanca to National Park Laguna Blanca (Argentina)	2024
Presented research and activities of the Castle Lake Research Station to the Mount Shasta community (California).	2020
Assisted in promoting undergraduate programs at the Universidad Nacional del Sur, Argentina)	2015-2023
Outreach activities at La Salada shallow lake about its importance to preservation and ecosystem Services Evaluation by the community	2014-2018

Selected Media Coverage & Interviews

Leader of the data management and software development of the Tahoe Environmental Observatory Network Link	2025
Leader of the software development of the Summit Environmental Watershed Observatory Network Link	2025
Participation in the University of Nevada, Reno Signs Historic Agreement with Summit Lake Paiute Tribe and Summit Lake Watershed Observatory Network video link	2025
Nevada Today. Microplastic Mayhem: How three researchers are analyzing particles in Lake Tahoe Link	2025
Journal Diario Jornada (Argentina). "CONICET research team monitors lakes to study algae and fish food sources". Link	2022
INQUA spotlight Early Research Career. The Spotlight #7	2022
paper highlighted : "Response of shallow lakes in the arid-semiarid Pampas of Argentina to Late Holocene hydroclimatic change (https://www.sciencedirect.com/science/article/pii/S1040618221004869#!)	
Environmental Monitor (U.S.). "Wildfire smoke alters a lake's ecology from the top to the bottom of the food chain". Link	2021

SELECTED CONFERENCE PRESENTATIONS

(Students, or technicians mentored or co-mentored by Seitz are underlined)

Ziaco E, Tromboni F, Scordo F, **Seitz C**, Chandra S. Long-term dynamics of forest and aquatic net primary productivity inferred from tree-ring and limnological records in a sub-alpine lake. EGU General Assembly 2025, 27 April- 2 May, Vienna, Austria.	2025
Tromboni F, Jativa C, **Seitz C**, Maasri A, Mohr S, Grossart HP, Grandi G, Bertuzzo E, Jähnig SC, Mendoza-Lera C, Lorke A, Cantonati M, Lupon A, Bernal S. Flow and Biodiversity Shifts Steer Riverine Ecosystem Metabolism: Insights from Experimental Stream Mesocosms. San Juan, Puerto Rico, May 18-22, at the Puerto Rico Convention Center. https://www.sfsannualmeeting.org/about	2025
Seitz C, Scordo F, Suenaga E, Carlson E, Chandra S. Spatial-Temporal variation of organic matter and nutrients in two mountains lakes. 2024 Mountain Climate Conference. September 16-19, Lake Tahoe, California. Oral Presentation and abstract.	2024
Mast E, Chandra S, Blaszczak J, Nussear KE, Suenaga E, Scordo F, **Seitz C.** Zoobenthic macroinvertebrate community responses to ice break up in subalpine lake habitats over 15 years (7855). ASLO 2024 Meeting 2–7 June 2024 Madison, Wisconsin, USA.	2024

Seitz C, Scordo F, Vélez MI, Perillo GME. El origen y morfología de los lagos pampeanos condicionan sus respuestas al cambio climático y antrópico. IV Congreso Iberoamericano de Limnología/ X Congreso Argentino de Limnología. July 31-August 4. Oral Presentation and abstract. 2023

Seitz C, Scordo F, Vélez MI, Cottet J, Villarrosa G. Análisis del efecto de la caída de tefras y su integración en la interpretación paleoecológica y paleoclimática. IV Congreso Iberoamericano de Limnología/ X Congreso Argentino de Limnología. July 31-August 4. Buenos Aires. Oral Presentation and abstract. 2023

Scordo F, **Seitz C**, Abasto B, Piccolo MC, Spetter CV, Perillo GME. Diferencias metabólicas y de limitación de nutrientes en la zona litoral-bentónica y pelágica de un lago oligotrófico de la Patagonia Argentina. IV Congreso Iberoamericano de Limnología/ X Congreso Argentino de Limnología. July 31-August 4. Oral Presentation and abstract 2023

Chandra S, Brahney J, Scordo F, **Seitz C**, Sadro S, Suenaga E, Carlson E, Tromboni F, Williamson C, Culpepper J, Rose K, Overholt E, Blasczak J, Loria K. Wildfire Smoke: Distribution, Particle Characterization, and Impacts on Lakes. ASLO Aquatic Sciences Meeting. 4–9 June. Palma de Mallorca, Spain. 2023

Tromboni F, Játiva-Guzmán C, Maasri A, Bernal S, Mohr S, Frank AH, Barth JAC, Dordoni M, Jähnig SC, Chandra C, **Seitz C**, Grossart HP. Effects of flow reduction and biodiversity decline on periphyton characteristics and carbon cycling. 36th Congress of the International Society of Limnology 7 – 10 August 2022. Oral Presentation and abstract. 2022

Suenaga E, Scordo F, Culpepper JA, Chandra S, **Seitz C**. Changes in Ice Cover Regimes Influence Zooplankton Community Structure in Subalpine Castle Lake, California. JASM 2022 Grand Rapids, Michigan, May 14-20. Oral Presentation and abstract. 2022

Seitz C, Vélez MI, Perillo GME. Environmental evolution of the lakes in the arid-semiarid Pampas of Argentina over the last millennium. IAL-IPA 2022 congress. San Carlos de Bariloche, Argentina. Oral Presentation and abstract. 2022

Seitz C, Vélez MI, Perillo GME. Drivers of ecological change in the pampean lakes during the last millennium. Pre-IAL IPA: ECR virtual meeting, April 26-29. Oral presentation. 2022

Seitz C, Scordo F, Suenaga E, Chandra S. Invasive Quagga Mussel: Potential Survival, Growth and Reproduction in a fresh to brackish water ecosystem. JASM 2022 Grand Rapids, Michigan, May 14-20. Oral Presentation and abstract 2022

Scordo F, Sadro S, Culpepper JA, **Seitz C**, Chandra S. Differential effects of wildfire smoke on habitat-specific pelagic and littoral lake metabolism. JASM 2022 Grand Rapids, Michigan, May 14-20. Oral Presentation and abstract 2022

San Pedro AP, Scordo F, **Seitz C**, Suenaga E, Chandra S. Contrasting Responses of Periphyton Growth to Nutrient Amendment across the Littoral-Benthic Habitat of a Mesotrophic Subalpine Lake (Castle Lake, CA). JASM 2022 Grand Rapids, Michigan, May 14-20. Oral Presentation and abstract. 2022

Chandra S, Scordo F, Suenaga E, Kelson S, Culpepper J, Sadro S, **Seitz C,** Rose K, Williams C, Tromboni F, Poulsen S, Scaff L, Fiorenza J, Caldwell T. Smoke on the Water! The Influence Regional Wildfires on the Ecological Structure of Castle Lake (CA, USA). 36th Congress of the International Society of Limnology, August 7 – 10. Abstract. 2022

Seitz C, Vélez MI, Perillo GME. Cambios en los regímenes de las lagunas pampeanas en el último milenio. XVII Reunión Argentina de Sedimentología y el VIII Congreso Latinoamericano de Sedimentología, Paraná, Argentina. Poster and abstract. 2021

Seitz C, Vélez MI, Perillo GME. Drivers of regime shift in the Pampean lakes: a paleolimnological perspective. ASLO-2020 Joint Meeting in Madison, Wisconsin.	2020
Vélez MI, Rusak JA, Conde D, Lozoya JP, **Seitz C**, Harmon T, Escobar Jaramillo JH, Perillo GME. Paleolimnological studies improve socio-ecological risk assessments: a tale of two lagoons. Poster. ASLO 2018 SUMMER MEETING, June 10-15, Victoria, BC, Canadá.	2018
Toniolo MA, **Seitz C,** Perillo GME. Origen y evolución de las depresiones mareales de una planicie de marea del estuario de Bahía Blanca, Argentina. XVI Reunión Argentina de Sedimentología, General Roca, Río Negro, Argentina.	2018
Seitz C, Vélez MI, Perillo GME. Inferencias Climáticas en base al registro sedimentario de la Laguna De Puan Durante El Holoceno. XVI Reunión Argentina de Sedimentología, General Roca, Río Negro, Argentina.	2018
Mavo Manstretta GM, Payares Peña NB, **Seitz C,** Vitale A, Perillo GME. Morfología de Fondo de un Meandro en un Canal de Marea del Estuario de Bahía Blanca. Poster. X Jornadas Nacionales de Ciencias del Mar. July 30 -Aug 10, Universidad de Buenos Aires, Buenos Aires, Argentina.	2018
Rodríguez Abaunza GA, López-Blanco C, Pérez L, **Seitz C,** Cuña Rodríguez C, Oxman B, Bennett K, Fontana S. Late-Holocene Paleoenvironmental Changes of Laguna Blanca Grande (Buenos Aires, Argentina) Inferred From Multi-Proxy Lacustrine Records. Reunião Regional Paleo Rs, Rio Grande, Brasil.	2017
Lavender TM, Scordo F, Perillo V, **Seitz C,** Brendel AS, Rusak JA, Arnott SE, Schamp BS, Wemple BC, Perillo GME. Aquatic ecosystem Services: Improving seasonal surface and ground water models for northern climates. Canadian Conference for Fisheries Research and Society of Canadian Limnologists (CCFFRS/SCL). January 5-8, Montreal, Canadá	2017
Lavender TM, Scordo F, Perillo V, **Seitz C,** Brendel AS, Rusak JA, Arnott SE, Wemple BC, Perillo GME, Piccolo MC, Caicedo MIV. "InVESTing in Aquatic Ecosystem Services: Improving conservation decision-making with the incorporation of a seasonal cryosphere in the modelling of water quantity" Poster GLEON 19, Mohunk Lake, New York, USA.	2017
Seitz C, Vítale JA, Perillo GME, Piccolo MC. "Bathymetry and morphometry of Pampean shallow lakes: an approach to the origin and productivity". Poster GLEON 18 meeting, July 4 – 8, Lunz and Gaming, Austria.	2016
Scordo F, **Seitz C,** Melo WD, Piccolo MC, Perillo GME. "Dinámica Geomorfológica De La Cuenca Del Río Senguer". Poster. 3ras Jornadas de Tecnologías de la Información Geográfica del Sur Argentino. August 24-26, Bahía Blanca.	2016
Scordo F, Perillo V, Lavander M, **Seitz C,** Brendel AS. "Preliminary results: Implementing the Seasonal Water Yield model (INVEST NAT-CAP) for three different watersheds across the Americas. The impact of GLEON fellowship program on SAFER student work experience". Poster GLEON 18 meeting. July 4 – 8, Lunz and Gaming, Austria.	2016
Seitz C, Vítale AJ, Perillo GME, Piccolo MC. Estudio geomorfológico de la laguna La Salada empleando un vehículo autónomo no tripulado. Poster. XXVII Reunión Científica de la Asociación Argentina de Geofísicos y Geodestas. November 10-14 San Juan, Argentina.	2014
Seitz C, Vítale AJ, Perillo GME, Piccolo MC. "Geomorphologic Study of La Salada Lake (Argentina) Employing Unmanned Autonomous Vehicle". Poster GLEON 16 meeting. Orford, Québec, Canadá, October 27- 31.	2014

Costilla PS, Bertoni B, **Seitz C,** Zapperi PA, Piccolo MC, Perillo GME. Initial progress of the SAFER education program about lake conservation and use. Poster GLEON 16 meeting, de Orford, Québec, Canadá, October 27- 31. 2014

Alfonso B, Bertoni B, Costilla P, Scordo F, **Seitz C**, Serna Zapata Y, Zapperi P, Rusak J. A SAFER integration of aquatic ecosystem risk assessment across the Americas. Poster. CNAES 2nd Annual Meeting & HQP Workshop. April 30-May 2, Montréal, Québec -Université de Québec à Montréal (UQAM) http://www.cnaes.ca/annual-meeting/ 2014

Alfonso B, Bertoni B, Costilla P, Scordo F, **Seitz C**, Serna Zapata Y, Zapperi P, Rusak JA. Student work in SAFER Project: an interdisciplinary study of aquatic ecosystem. Poster. Muskoka Summit on Environment: Environment vs Economy. Solving the Dichotomy. May 8-9. Bracebridge, Ontario. http://muskokasummit.org/2014-summit/ 2014